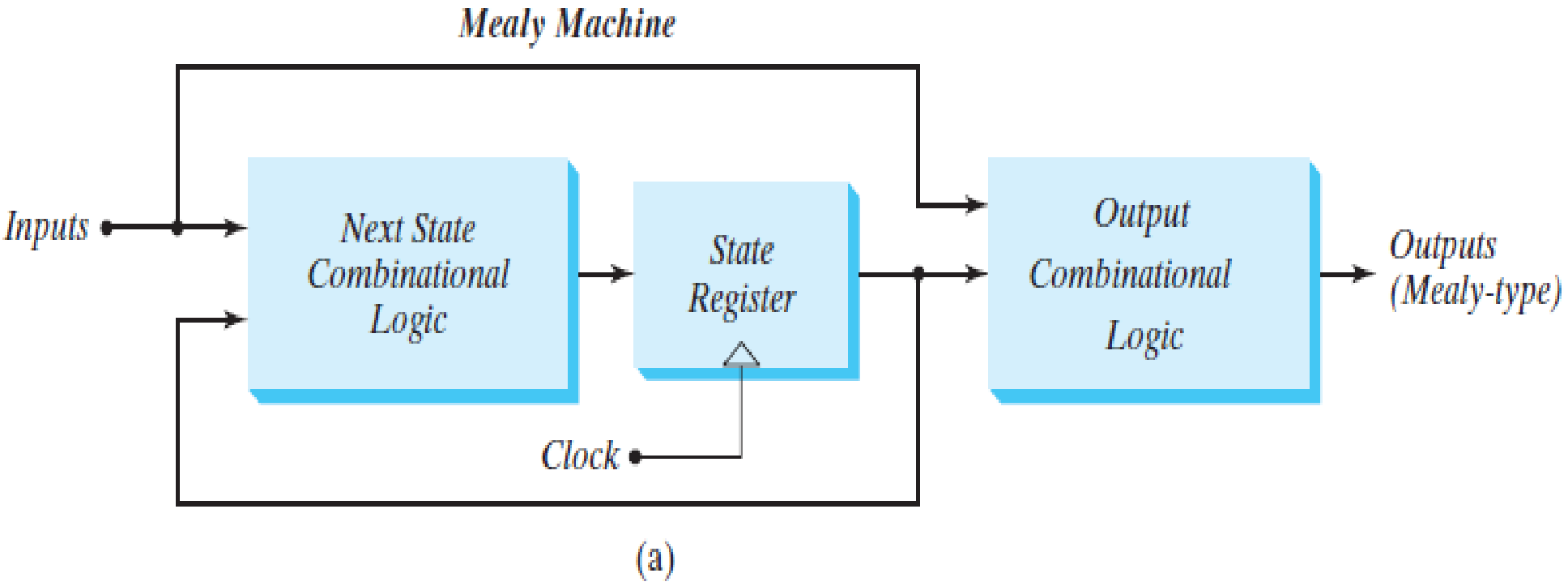
Mealy and Moore Models of Finite State Machines

The most general model of a sequential circuit has inputs, outputs, and internal states. It is customary to distinguish between two models of sequential circuits: the Mealy model and the Moore model. They differ only in the way the output is generated.

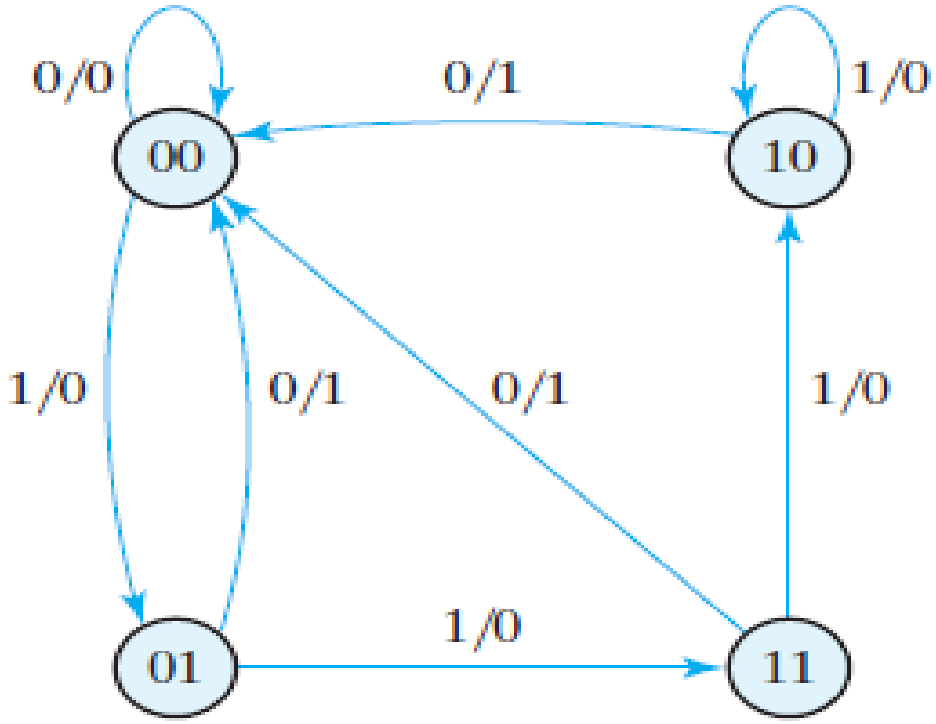
Mealy Model



(a)

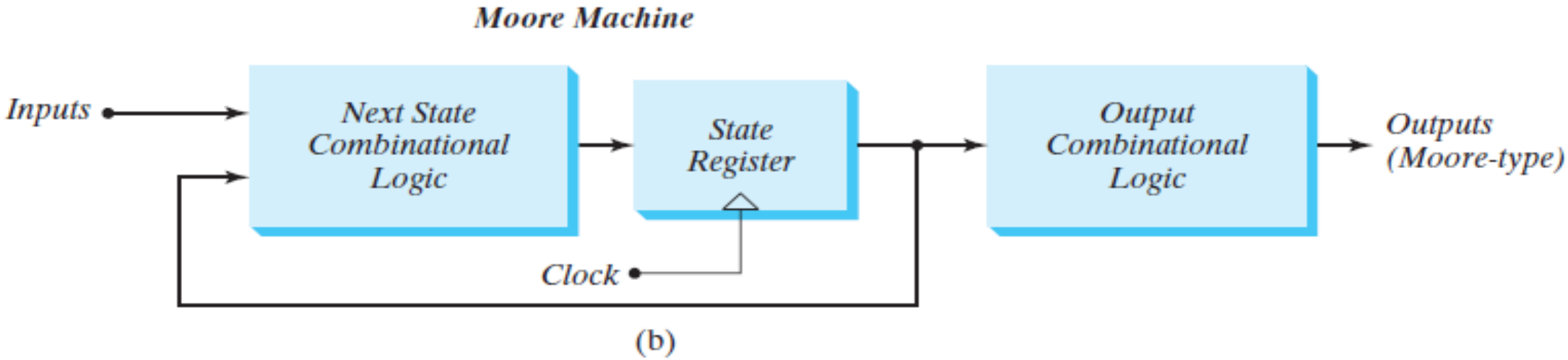
In the Mealy model, the output is a function of both the present state and the input.

Example:
State Diagram:



Both the input and output values are shown, separated by a slash along the directed lines between the states.

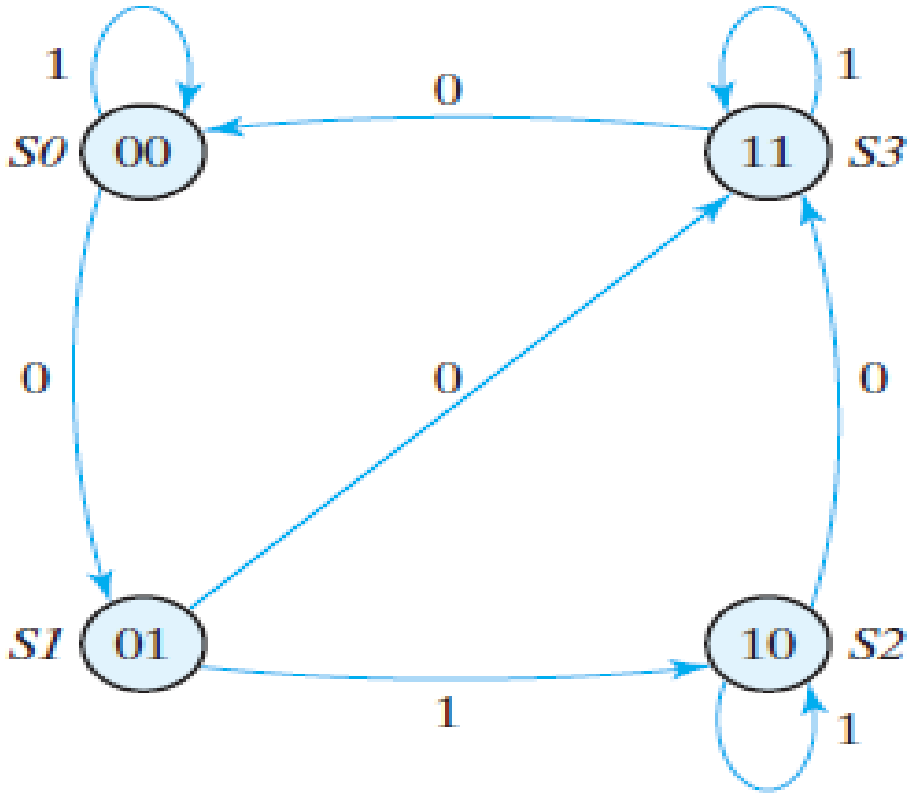
Moore Model



- In the Moore model, the output is a function of only the present state.
- A circuit may have both types of outputs.
- The two models of a sequential circuit are commonly referred to as a finite state machine, abbreviated FSM.

Example:

State Diagram:



It has only inputs marked along the directed lines. The outputs are the flip-flop states marked inside the circles. There may be other outputs also, which are functions of present state only.

Design a circuit that meets the following specification:

1. The circuit has one input, *w*, and one output, *z*.
2. All changes in the circuit occur on the positive edge of the clock signal.
3. The output *z* is equal to 1 if during two immediately preceding clock cycles the input *w* was equal to 1. Otherwise, the value of *z* is equal to 0.

From this specification it is apparent that the output *z* depends on both present and past values of *w*.

Consider the sequence of values of the *w* and *z* signals for the clock cycles shown in Figure 6.2.

Clock cycle:	t_0	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8	t_9	t_{10}
w:	0	1	0	1	1	0	1	1	1	0	1
z:	0	0	0	0	0	1	0	0	1	1	0

Figure 6.2 Sequences of input and output signals.

As seen in the figure, for a given value of input *w* the output *z* may be either 0 or 1. For example, $w = 0$ during clock cycles $t2$ and $t5$, but $z = 0$ during $t2$ and $z = 1$ during $t5$. Similarly, $w = 1$ during $t1$ and $t8$, but $z = 0$ during $t1$ and $z = 1$ during $t8$. This means that *z* is not determined only by the present value of *w*, so there must exist different states in the circuit that determine the value of *z*.

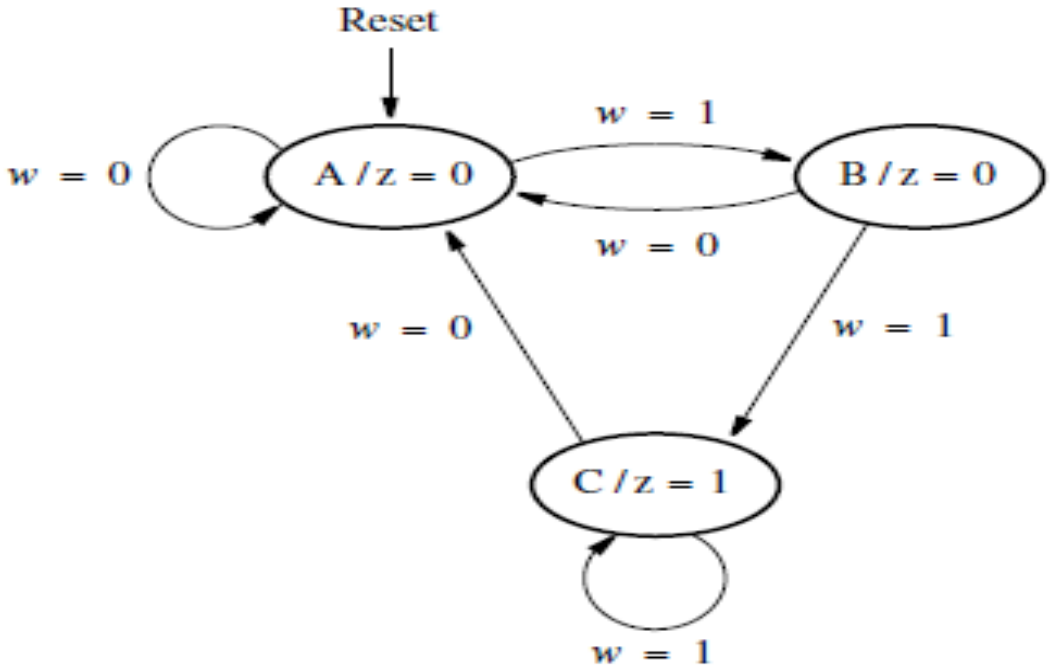


Figure 6.3 State diagram of a simple sequential circuit.

Present state	Next state		Output
	$w = 0$	$w = 1$	z
A	A	B	0
B	A	C	0
C	A	C	1

Figure 6.4 State table corresponding to Figure 6.3.

Reset input is used to force the circuit into state *A*, regardless of what state the circuit happens to be in.

	Present state	Next state		Output
		$w = 0$	$w = 1$	z
	y_2y_1	Y_2Y_1	Y_2Y_1	
A	00	00	01	0
B	01	00	10	0
C	10	00	10	1
	11	dd	dd	d

Figure 6.6 State-assigned table corresponding to Figure 6.4.

$$Y_1 = w\overline{y}_1\overline{y}_2$$
$$Y_2 = w(y_1 + y_2)$$
$$z = y_2$$

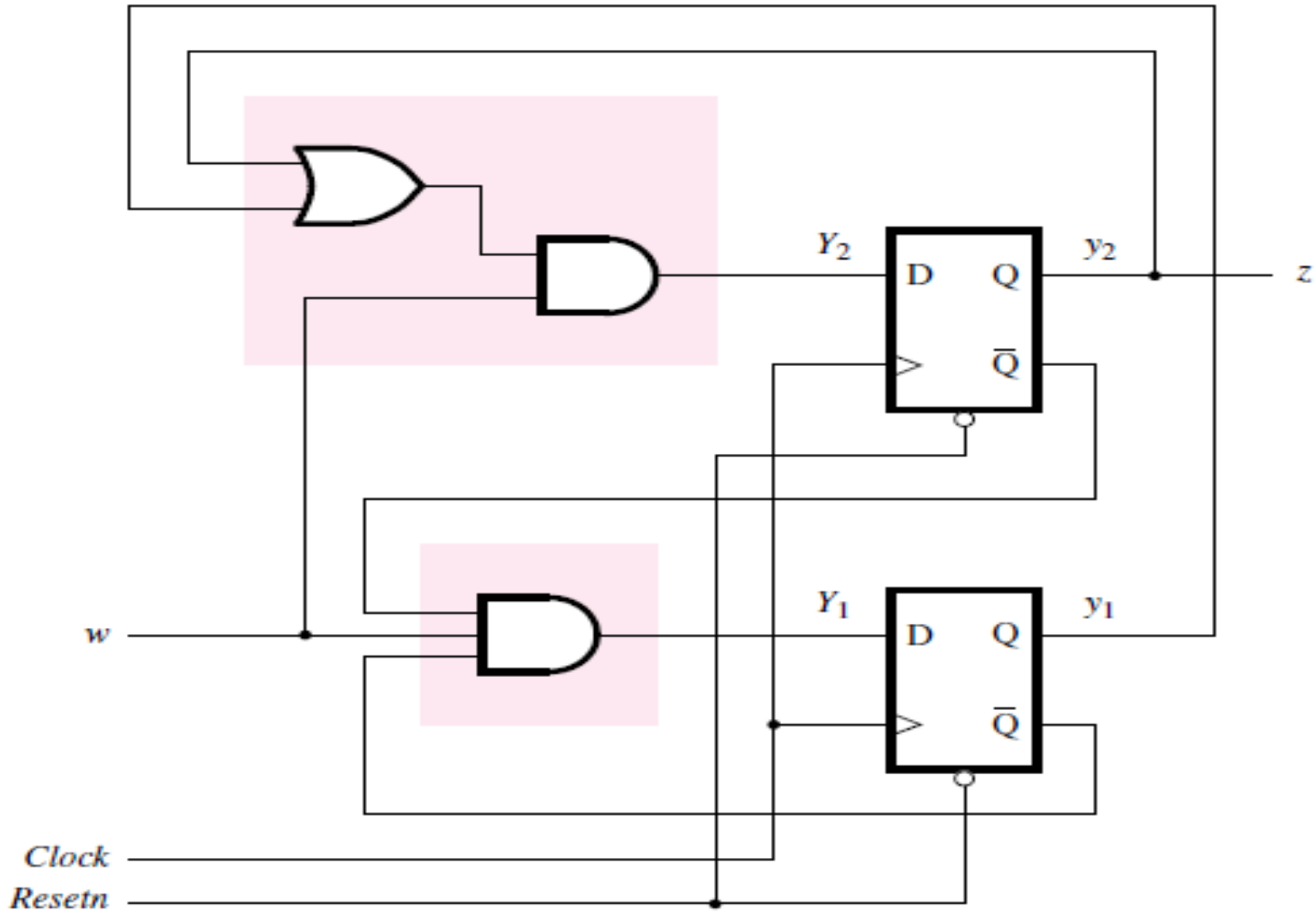
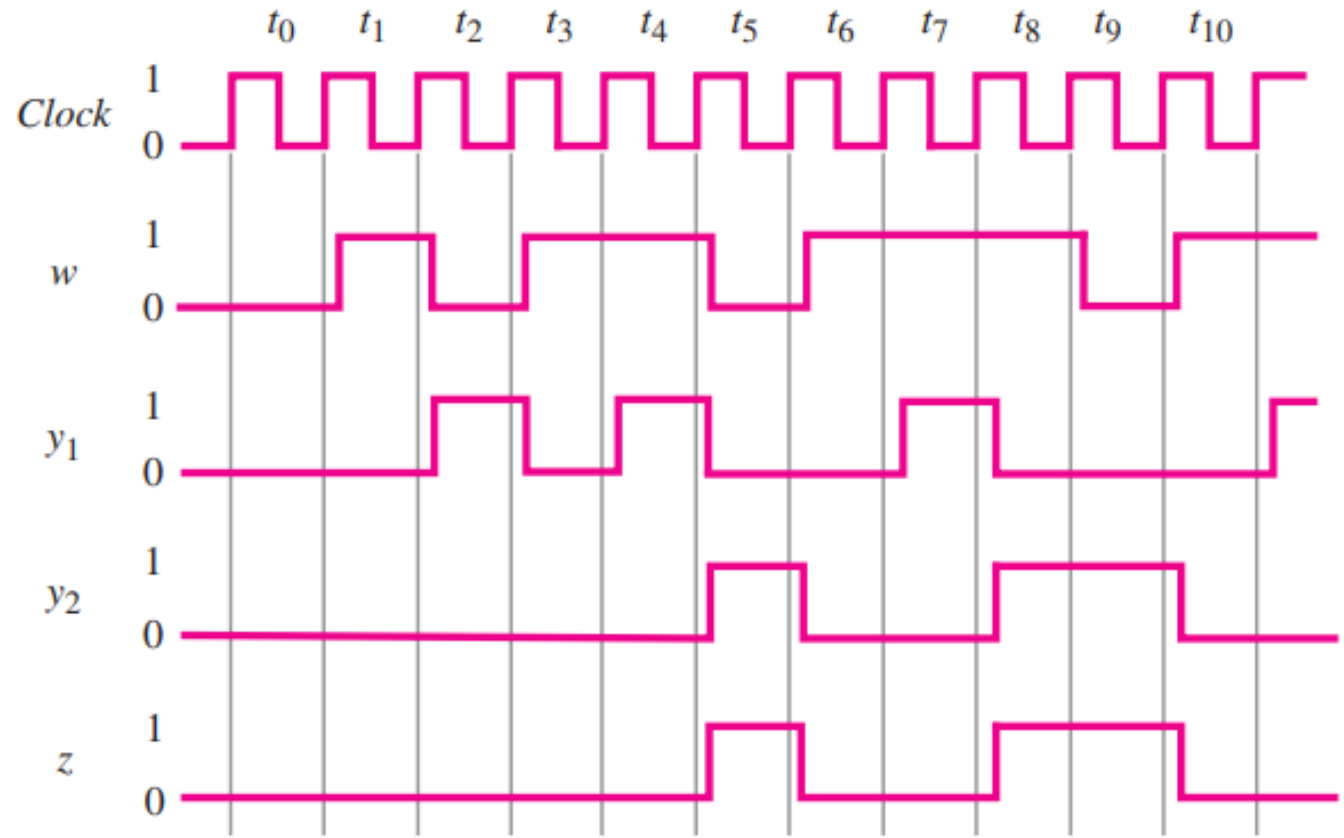


Figure 6.8 Final implementation of the sequential circuit.

Clock cycle:	t_0	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8	t_9	t_{10}
w:	0	1	0	1	1	0	1	1	1	0	1
z:	0	0	0	0	0	1	0	0	1	1	0



STATE REDUCTION

A more practical way to determine state equivalence uses the state equivalence theorem:

- $s_i \equiv s_j$ if and only if for every single input X, the outputs are the same and the next states are equivalent.
- When using the definition of equivalence, we must consider all input sequences, but we do not need any information about the internal state of the system.
- When using the state equivalence theorem, we must look at both the output and next state, but we need to consider only single inputs rather than input sequences

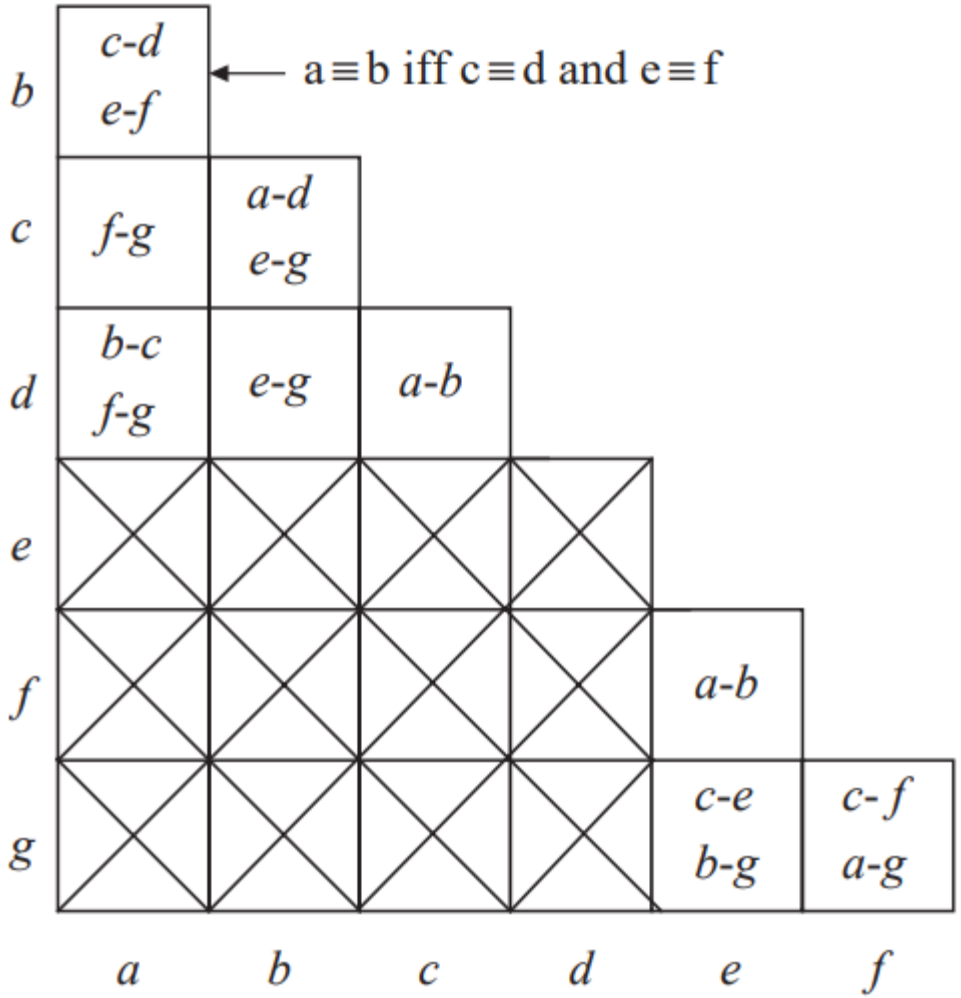
Example

PS	Next State		Output Z	
	X=0	X=1	X=0	X=1
a	c	f	0	0
b	d	e	0	0
c	h	g	0	0
d	b	g	0	0
e	e	b	0	1
f	f	a	0	1
g	c	g	0	1
h	c	f	0	0

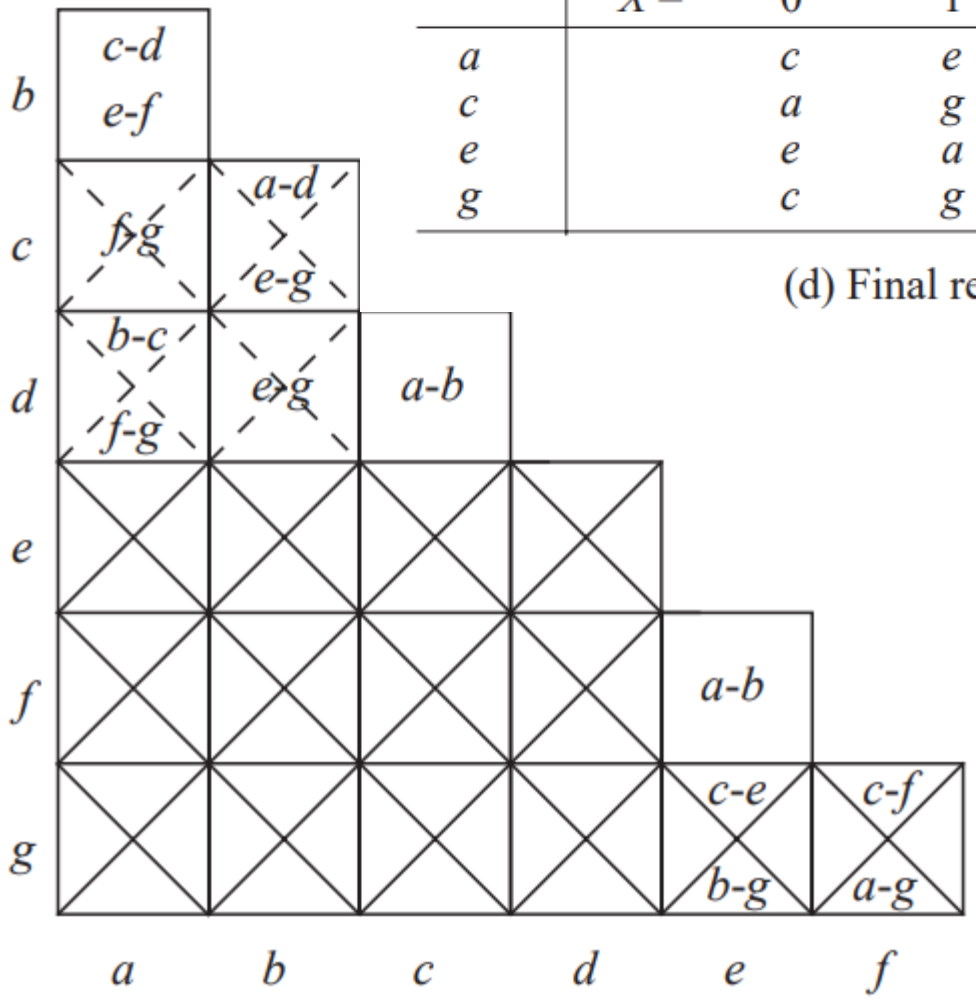
- The table can be reduced by eliminating equivalent states.
- First, observe that states a and h have the same next states and outputs when X = 0 and also when X= 1.
- since the outputs for states a and b are the same, a ≡ b if and only if c ≡ d and e ≡ f. We say that c-d and e-f are implied pairs for a-b.
- Since the outputs for states a and b are the same, a= b if and only if c ≡ d and e ≡ f. We say that c-d and e-f are implied pairs for a-b.
- To keep track of the implied pairs, we make an implication chart

Present State	Next State $X=0$	1	Present Output $X=0$	1
a	*c*	*f*	0	0
b	*d*	*e*	0	0
c	~~*h*~~*a*	*g*	0	0
d	*b*	*g*	0	0
e	*e*	*b*	0	1
f	*f*	*a*	0	1
g	*c*	*g*	0	1
~~*h*~~	~~*c*~~	~~*f*~~	~~0~~	~~0~~

Implication chart



(b) Implication chart (first pass)



(c) After second and third passes

	$X=$	0	1	$X=$	0	1
a		*c*	*e*		0	0
c		*a*	*g*		0	0
e		*e*	*a*		0	1
g		*c*	*g*		0	1

(d) Final reduced table

BCD to Excess-3 Conversion

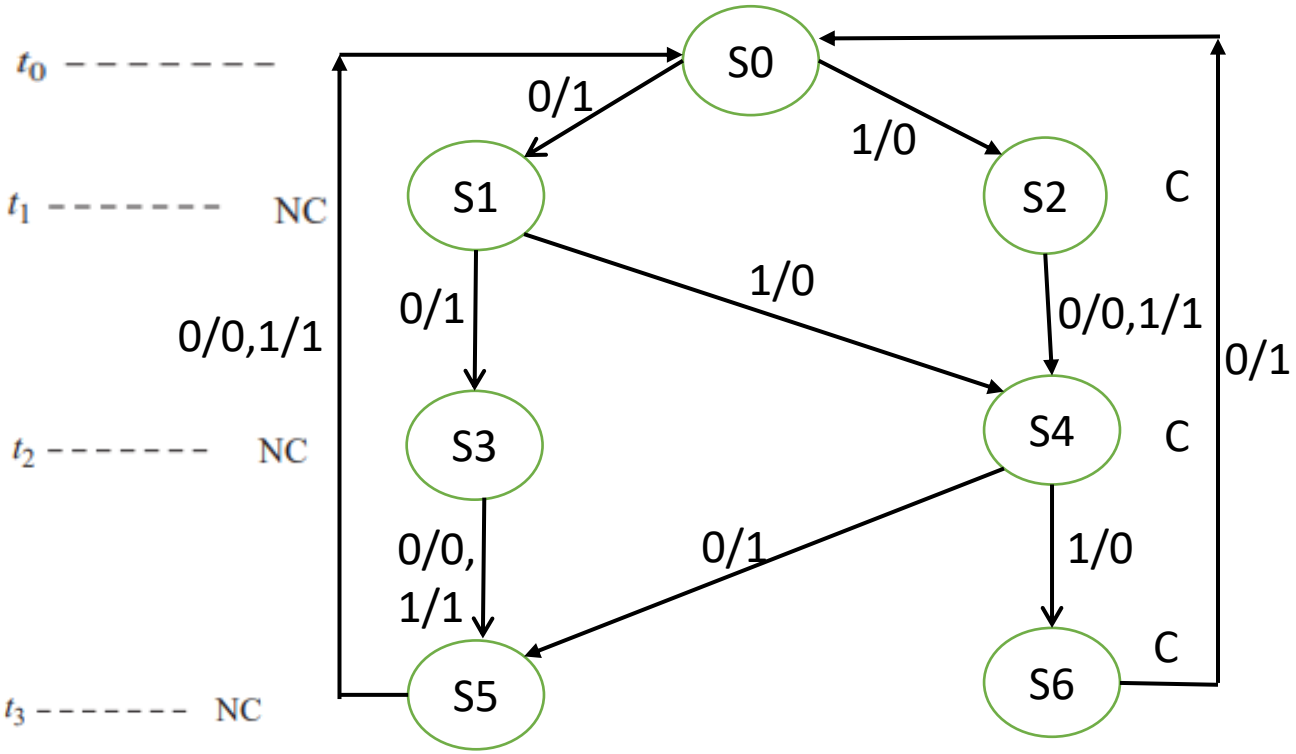
X Input (BCD)				Z Output (excess-3)			
t_3	t_2	t_1	t_0	t_3	t_2	t_1	t_0
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0

$$\begin{array}{r} 0\ 1\ 0\ 0 \\ +\ 0\ 0\ 1\ 1 \\ \hline 0\ 1\ 1\ 1 \end{array}$$

$$\begin{array}{r} 0\ 1\ 0\ 1 \\ +\ 0\ 0\ 1\ 1 \\ \hline 1\ 0\ 0\ 0 \end{array}$$

	NS		Z	
PS	$X=0$	$X=1$	$X=0$	$X=1$
S0	S1	S2	1	0
S1	S3	S4	1	0
S2	S4	S4	0	1
S3	S5	S5	0	1
S4	S5	S6	1	0
S5	S0	S0	0	1
S6	S0	–	1	–

(b) State table



STATE ASSIGNMENT

I. States that have the same next state (NS) for a given input should be given adjacent assignments (look at the columns of the state table).

II. States that are the next states of the same state should be given adjacent assignments (look at the rows).

III. States that have the same output for a given input should be given adjacent assignments.

	NS		Z	
PS	$X=0$	$X=1$	$X=0$	$X=1$
S0	S1	S2	1	0
S1	S3	S4	1	0
S2	S4	S4	0	1
S3	S5	S5	0	1
S4	S5	S6	1	0
S5	S0	S0	0	1
S6	S0	–	1	–

(b) State table

I. (1, 2), (3, 4), (5, 6)

II. (1, 2), (3, 4), (5, 6)

III. (0, 1, 4, 6), (2, 3, 5)

Q_2Q_3 \ Q_1	0	1
00	S0	S1
01		S2
11	S5	S3
10	S6	S4

(a) Assignment map

(in the $X = 1$ column, S_1 and S_2 both have NS S_4; in the $X = 0$ column, S_3 and S_4 have NS S_5, and S_5 and S_6 have NS S_0)
(S_1 and S_2 are NS of S_0; S_3 and S_4 are NS of S_1; and S_5 and S_6 are NS of S_4)

	$Q_1^+ Q_2^+ Q_3^+$		Z	
$Q_1Q_2Q_3$	$X=0$	$X=1$	$X=0$	$X=1$
000	100	101	1	0
100	111	110	1	0
101	110	110	0	1
111	011	011	0	1
110	011	010	1	0
011	000	000	0	1
010	000	xxx	1	x
001	xxx	xxx	x	x

(b) Transition table

Q_2Q_3 \ XQ_1	00	01	11	10
00	1	1	1	1
01	X	1	1	X
11	0	0	0	0
10	0	0	0	X

$$D_1 = Q_1^+ = Q_2'$$

Q_2Q_3 \ XQ_1	00	01	11	10
00	0	1	1	0
01	X	1	1	X
11	0	1	1	0
10	0	1	1	X

$$D_2 = Q_2^+ = Q_1$$

Q_2Q_3 \ XQ_1	00	01	11	10
00	0	1	0	1
01	X	0	0	X
11	0	1	1	0
10	0	1	0	X

$$D_3 = Q_3^+ = Q_1Q_2Q_3 + X'Q_1Q_3' + XQ_1'Q_2'$$

Q_2Q_3 \ XQ_1	00	01	11	10
00	1	1	0	0
01	X	0	1	X
11	0	0	1	1
10	1	1	0	X

$$Z = X'Q_3' + XQ_3$$

